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Supplemental information

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Supporting Information

Insights into Biofilm Architecture and Maturation Enable Improved Clinical Strategies for Exopolysaccharide-Targeting Therapeutics

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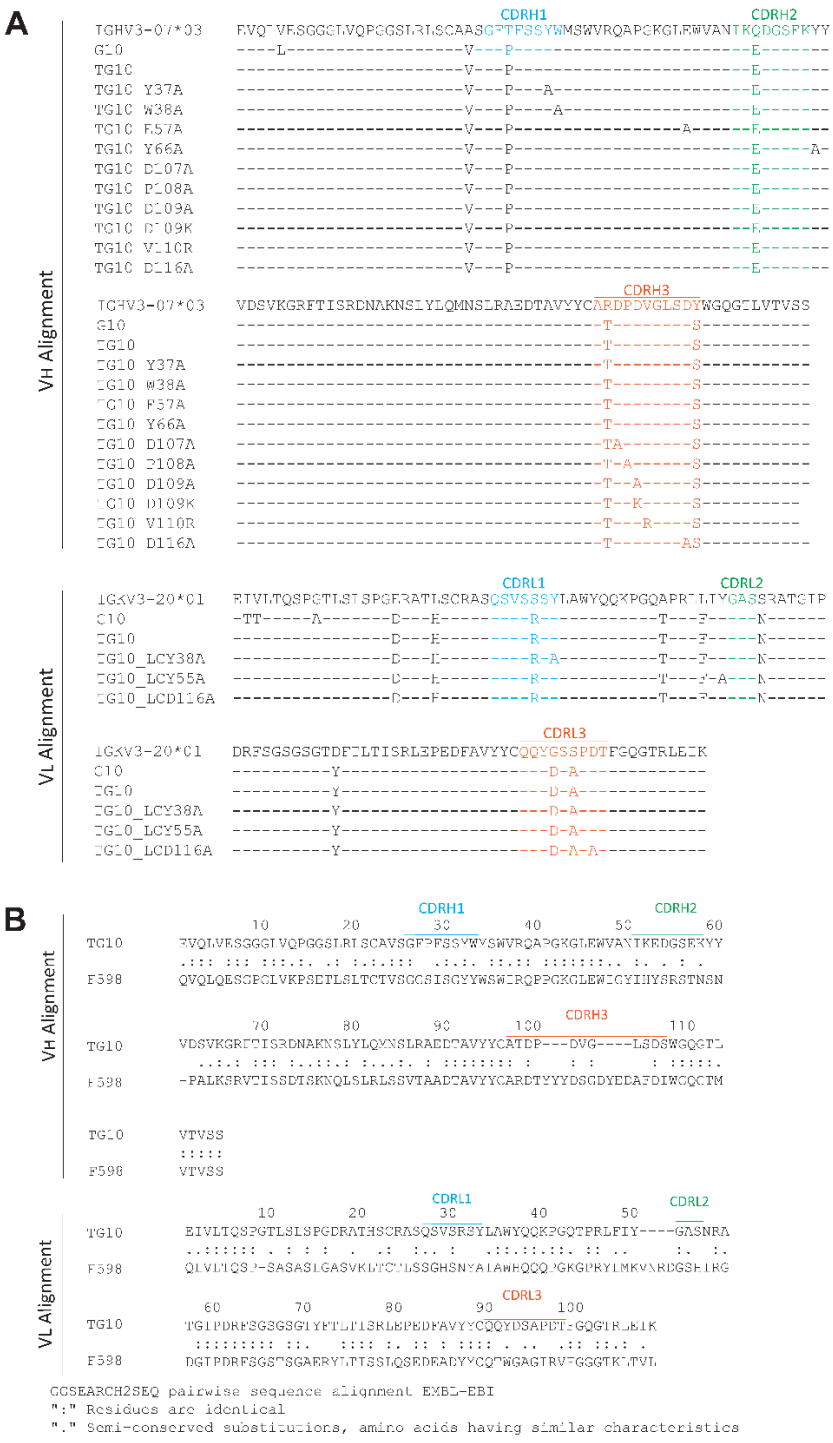


Figure S1. Sequence and Structural analysis for TG10. Related to Figure 3. (**A**) TG10 iGl, TG10, and TG10 mutant sequences. Rare mutations for the heavy chain: T29P, Q58E. Rare mutations for the light chain: L21H, L53F, D86Y. (**B**) TG10 and F598 sequence alignment. Aligned sequences show significant differences in the CDRs of TG10 and F598. [1,2]

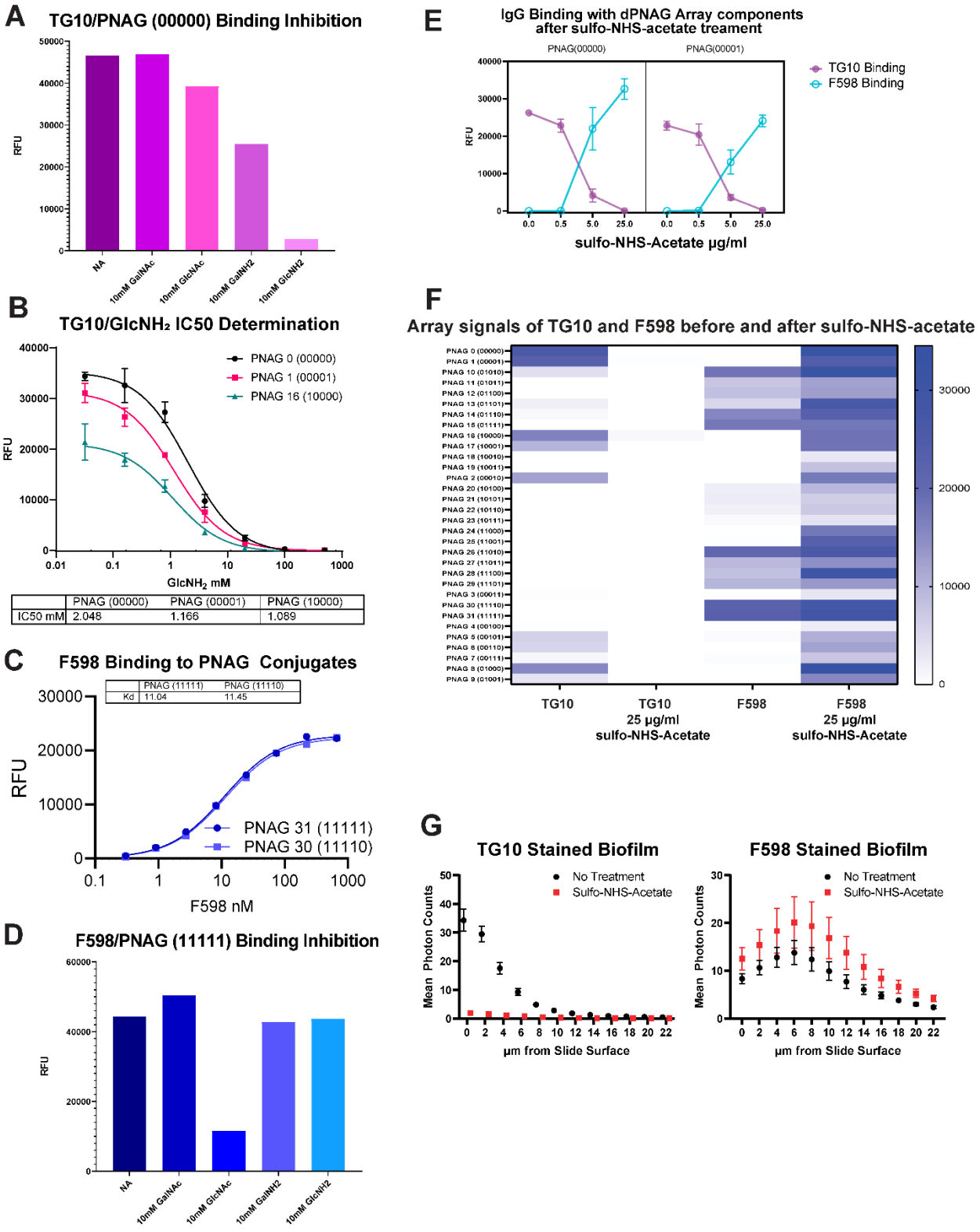


Figure S2. Binding of TG10 and F598 on the NGP microarray. Related to Figure 2 (**A**) Small molecule inhibition of TG10 with PNAG 0(00000). (**B**) TG10/dPNAG inhibition by $GlcNH_2$. IC50 determined for PNAG 0(00000), PNAG 1(00001) and PNAG 16(10000). (**C**) Apparent kDs of F598 with array

components PNAG 31(11111) and PNAG 30(00001). App. kD determined on NGP microarray. (**D**) Small molecule inhibition of F598 with PNAG 31(11111). (**E**) Effects of chemical acetylation on anti-PIA mAb binding. Sulfo-NHS-acetate reduces binding of TG10 to dPNAG conjugates PNAG(00000) and PNAG(00001) in a concentration-dependent manner. Conversely, sulfo-NHS-acetate effectively turns on F598 binding to dPNAG conjugates PNAG(00000) and PNAG(00001) in a concentration-dependent manner. (**F**) RFU heatmap of anti-PNAG mAbs on neoglycoprotein microarray with and without 25 mM sulfo-NHS-acetate. Chemical acetylation abolishes TG10 binding to all array components while chemical acetylation greatly increases F598 binding to all PNAG array components. (**G**) TG10 biofilm binding eliminated following sulfo-NHS-acetate treatment. F598 binding is statistically significantly higher following sulfo-NHS-acetate treatment. Seven X/Y independent z-stack images were acquired in 1 μm depth intervals, and the mean photon count with SEM was calculated in ImageJ.

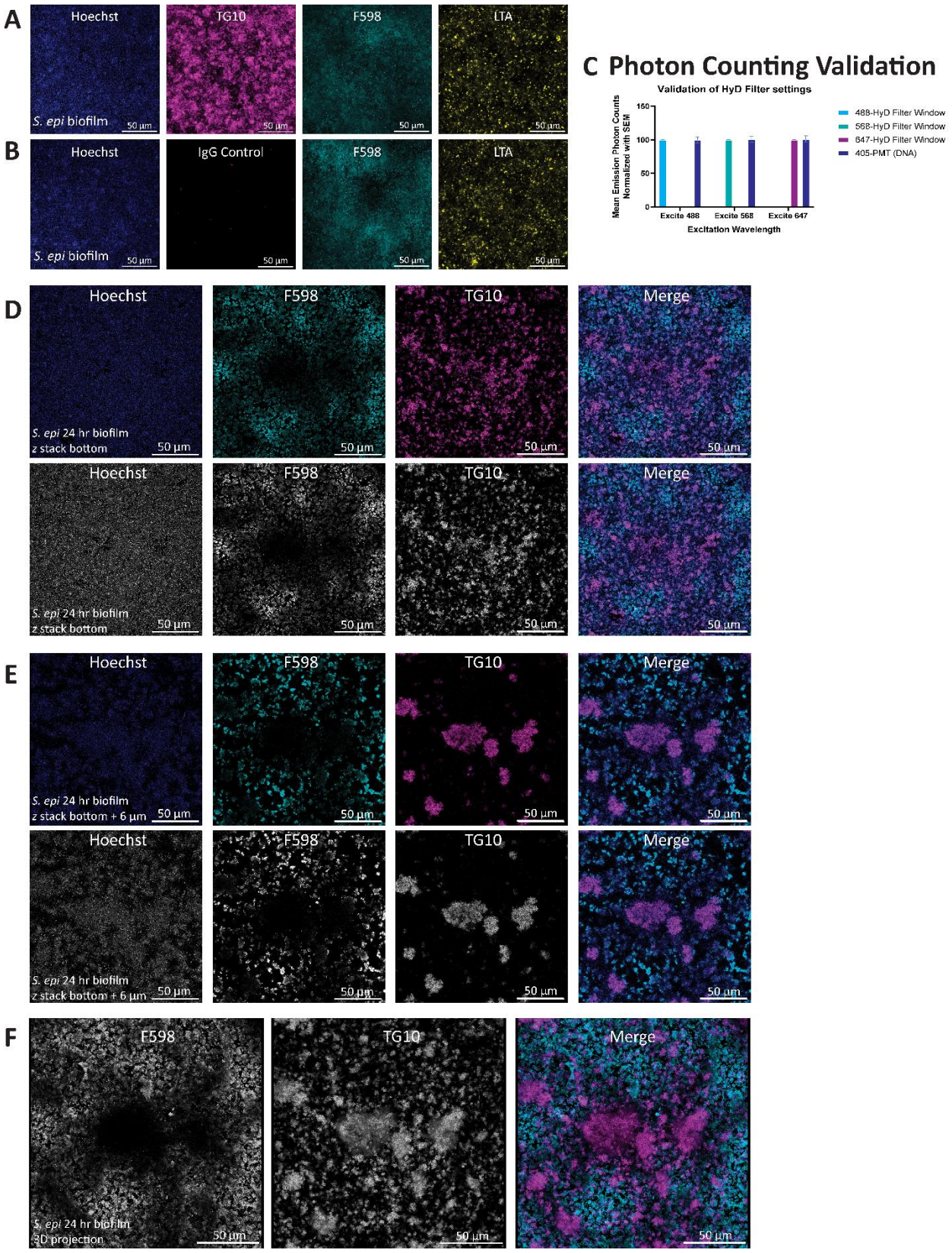

A
Hoechst
TG10
F598
LTA
S. epi biofilm
50 µm
B
Hoechst
IgG Control
F598
LTA
S. epi biofilm
50 µm
C Photon Counting Validation
Validation of HyD Filter settings
Mean Emission Photon Counts Normalized with SEM
150
100
50
0
Excite 488
Excite 568
Excite 647
Excitation Wavelength
488-HyD Filter Window
568-HyD Filter Window
647-HyD Filter Window
405-PMT (DNA)
D
Hoechst
F598
TG10
Merge
S. epi 24 hr biofilm
z stack bottom
50 µm
E
Hoechst
F598
TG10
Merge
S. epi 24 hr biofilm
z stack bottom + 6 µm
50 µm
F
F598
TG10
Merge
S. epi 24 hr biofilm
3D projection
50 µm

Figure S3. TG10-biofilm binding relative to IgG control (trastuzumab) and representative images and grayscale images of *S. epi* biofilms fixed and stained with TG10-647 (magenta), F598-488 (cyan), and Hoechst 33342 (blue). Related to Figure 4. (**A**) Biofilm stained with TG10. Hoechst (blue), TG10 (magenta), F598 (cyan), and anti-LTA (yellow). (**B**) Biofilm stained with IgG Control. Hoechst (blue), trastuzumab (magenta), F598 (cyan), and anti-LTA (yellow). (**C**) Emission photo counts at each filter used in the experiments demonstrate little to no noise/signal carryover. (**D**) Representative color and grayscale images of TG10 localizing to *S. epidermidis* at slide surface in the biofilm. (**E**) Representative color and grayscale images of TG10 localizing to *S. epidermidis* at 6 µm above the basement layer/slide surface. F598 binding is focused on the surface of these microcolonies. Scale bar 50 µm. (**F**) 3D projections grayscale and color merge of a biofilm Z-stack representing 200 slices. Scale bars 50 µm.

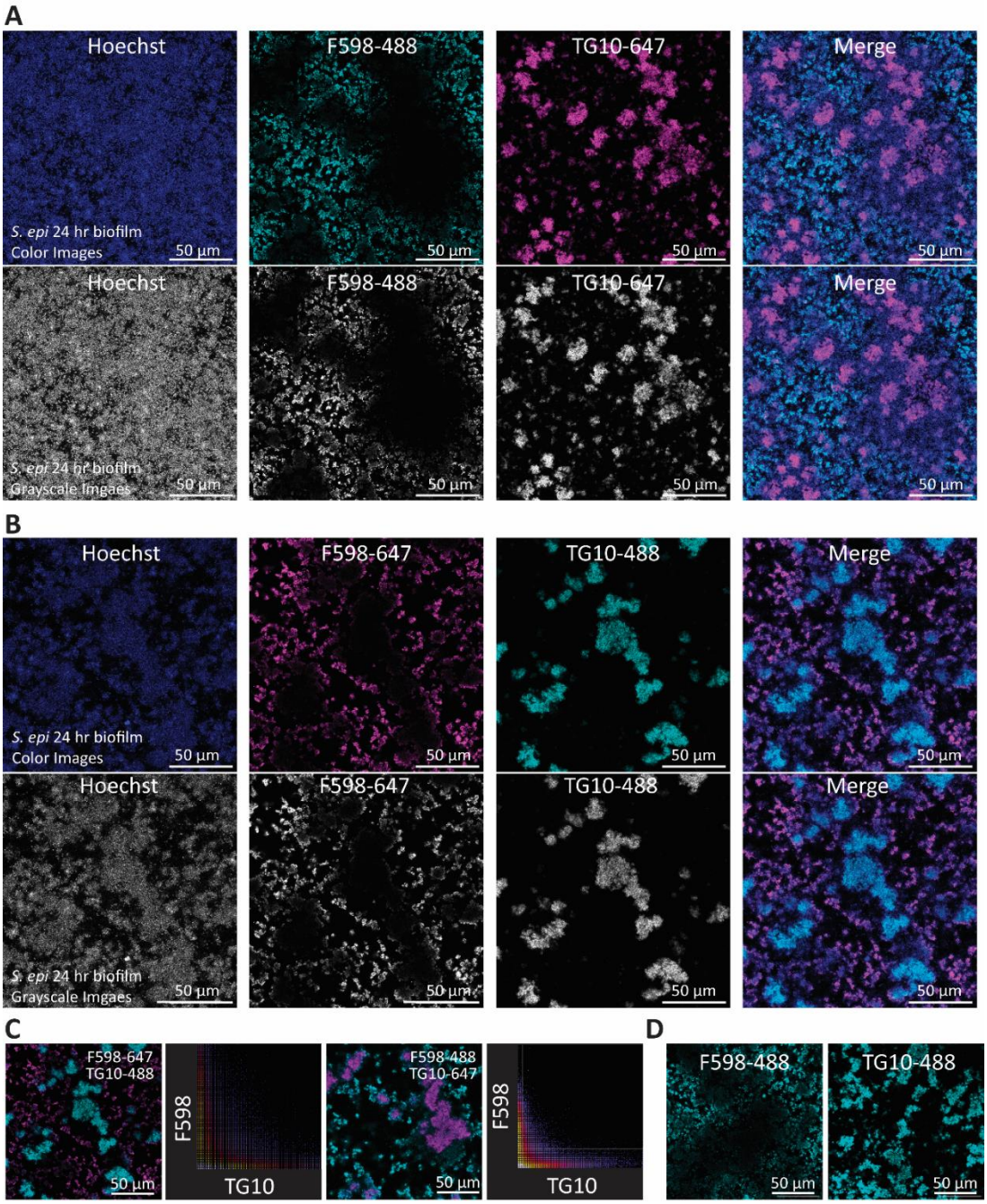


Figure S4. Fluorescent labels do not influence antibody localization. Related to Figure 4. (**A**) Hoechst (blue), F598-AF488 (cyan), TG10-AF647 (magenta), and merged image with corresponding grayscale images below. (**B**) Hoechst (blue), F598-AF647 (magenta), TG10-AF488 (cyan), and merged image with corresponding grayscale images below. (**C**) Coloc 2 algorithm in ImageJ shows very little colocalization or positive correlation regardless of fluorescent dye combination. (**D**) Labeling F598 or TG10 with the same dye (AF488) and staining biofilms with a single mAb consistently yields TG10 localizing to microcolonies and F598 localizing to areas surround bulk biofilm material. Scale bar 50 µm.

Supplementary References

(1) Shehata, L.; Maurer, D. P.; Wec, A. Z.; Lilov, A.; Champney, E.; Sun, T.; Archambault, K.; Burnina, I.; Lynaugh, H.; Zhi, X.; et al. Affinity Maturation Enhances Antibody Specificity but Compromises Conformational Stability. *Cell Reports* **2019**, *28* (13), 3300-3308.e3304, Article. DOI: 10.1016/j.celrep.2019.08.056 Scopus. Temme, J. S.; Crainic, J. A.; Walker, L. M.; Yang, W.; Tan, Z.; Huang, X.; Gildersleeve, J. C. Microarray-guided evaluation of the frequency, B cell origins, and selectivity of human anti-glycan antibodies reveals new insights and novel antibodies. *J. Biol. Chem.* **2022**, *298*, 102468.

(2) Dunbar, J.; Krawczyk, K.; Leem, J.; Marks, C.; Nowak, J.; Regep, C.; Georges, G.; Kelm, S.; Popovic, B.; Deane, C. M. SAbPred: a structure-based antibody prediction server. *Nucleic Acids Res* **2016**, *44* (W1), W474-478. DOI: 10.1093/nar/gkw361 From NLM. Raybould, M. I. J.; Marks, C.; Krawczyk, K.; Taddese, B.; Nowak, J.; Lewis, A. P.; Bujotzek, A.; Shi, J.; Deane, C. M. Five computational developability guidelines for therapeutic antibody profiling. *Proc Natl Acad Sci U S A* **2019**, *116* (10), 4025-4030. DOI: 10.1073/pnas.1810576116 From NLM.